Intelligent Commodity Investing: Opportunities and Challenges

By Hilary Till
Principal, Premia Research LLC; and
Research Associate, EDHEC-Risk Institute

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This paper will discuss the historical underpinning of the current boom in commodity prices; and alert the busy reader to some unexpected pitfalls when investing in this theme. We will conclude with some observations on how to potentially take advantage of this asset class' opportunities.

According to Bannister (2006, 2007) over the very long term, commodity producers periodically need sufficient pricing to replenish physical and human assets. "Every 15 to 20 years, farmers, miners, [and] drillers … [as well as] developing countries (that [are] tied to the land) must receive pricing power to replenish fixed assets and draw a new generation into the business, only [later] to be smashed against the rocks in a deflation for 15 or 20 years until the workers and assets are retired." In this harsh cycle, "commodities (and firms that produce commodities or service the needs of commodity producers) periodically out-perform the S&P 500 for decade-and-a-half cycles. One such cycle began 8 years ago."

During the current boom, the macro case for commodity investments has relied on the following three factors:

(1) Adverse supply shocks resulting from the aging energy infrastructure in the U.S. and Europe;

(2) expanding Asian demand, particularly from China; and

(3) as a dollar hedge.

While there may be emerging oil scarcity, at least in politically stable parts of the world, is that the whole story on the oil rally? Since the founding of the Federal Reserve Board in 1913, it appears that money-supply growth has driven supply-inelastic crude-oil prices, notes Bannister.

When we examine the period, December 31st, 2001 through December 31st, 2007, we find that front-month crude oil prices have increased by about 380% in dollar terms. In view of Bannister's century-long analysis, one might wonder how much of the current price rise is due to oil scarcity, and how much is due to monetary debasement.

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If one denominates oil in say, fractional ounces of gold rather than in dollars, one comes away with a rather different picture. Using gold as the "unit of account," crude has instead rallied only 61% over the six-year period from the end of 2001 to the end of 2007.

"In an environment of monetary debasement … [in which] cash rapidly loses … its purchasing power, all goods, services, and … assets become currencies as investors … [seek out ways] to protect the purchasing power of their … [savings], as discussed by Faber (2007).

The main issue with choosing other stores-of-value besides cash is whether the cost-of-carry for these investments is unacceptably high. This is a key question in deciding upon whether commodity futures investments are suitable for an investor.

For example, the Goldman Sachs Commodity Energy Excess Return subindex lost -30% in 2006, largely because of the steep contangos in the crude oil futures market at the time. This is plainly too high a cost for a portfolio hedge.

Because of the volatility of crude oil prices and the potentially prohibitive cost-of-carry for an investment focused on crude-oil futures prices, energy and commodity investors had been drawn to relative-value energy hedge funds, especially prior to 2007.

As discussed in the 2007 Risk Book, *Intelligent Commodity Investing*, there are indeed potentially profitable spreading opportunities around build/draw cycles in commodity inventories. These opportunities tend to be monetized through calendar spreads and processor-margin spreads. But even with these strategies, there have been a number of challenges in profiting from energy relative-value trading due to frequent structural breaks in historical relationships. This was particularly the case during September 2006 with the Amaranth debacle, which was explained in Till (2006).

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Another recent risk concern has been how correlated the commodity markets have become with other risky assets, at least over short time horizons. Commodities were clearly not immune from sharp episodes of widespread deleveraging of risky investments during the past two years.

One example of simultaneous deleveraging is from February 27th, 2007. At the end-of-the-trading day, one saw algorithmic strategies simultaneously deleverage across numerous risky investments, including in popular commodity plays.

This phenomenon, again, became of concern on August 16th of 2007, the day before the Federal Reserve Board cut the discount rate. On that date, *all* commodity markets in the Dow Jones AIG Commodity Index were down, as shown in Figure 1, along with *all* other risky assets.

Figure 1

Intraday Performance of Commodities Within the Dow Jones AIG (DJAIG) Commodity Index On August 16, 2007

DJAIG MOVERS 8/16/2007 10:07am CST				
Commodity		Price	Change	% Change
LMAHDS03	Aluminum	2543.00y	-9.00	-0.35
NGX7	Natural Gas	7.791	-0.046	-0.59
W Z7	Wheat	688 3/4	-8 1/4	-1.18
LCV7	Live Cattle	94.600	-1.325	-1.38
LHV7	Lean Hogs	67.550	-1.025	-1.49
LMZSDS03	Zinc	3230.00y	-65.00	-1.97
XBX7	RBOB Gasoline	187.43	-3.95	-2.06
GCZ7	Gold	665.20	-14.50	-2.13
CTZ7	Cotton	58.85	-1.33	-2.21
CLX7	Crude Oil	71.10	-1.73	-2.38
HOX7	Heating Oil	201.55	-4.99	-2.42
C Z7	Corn	336 1/2	-8 3/4	-2.53
LMNIDS03	Nickel	26500.0y	-800.0	-2.93
SBV7	Sugar	9.16	-0.29	-3.07
KCZ7	Coffee	119.30	-3.90	-3.17
BOZ7	Soybean Oil	35.27	-1.25	-3.42
SIZ7	Silver	12.290	-0.445	-3.49
S X7	Soybeans	821	-33 1/2	-3.92
HGZ7	Copper	314.80	-17.40	-5.24

Source: Bloomberg

More recently, during the week of March 17th, 2008, market participants appeared to embrace a “preservation-of-capital” stance in the aftermath of the near collapse of Bear Stearns. Not only did three-month U.S. Treasury Bills hit a nadir of 39 bps in (annualized) yield, but the commodity markets witnessed a weekly sell-off, the scale of which had not been seen since 1956, according to Carpenter and Munshi (2008).

How should investors be positioned going forward, even acknowledging that over short time horizons, the fortunes of commodity prices may be *strongly* related to other risk assets?

The current function of (light sweet) crude oil prices is to search for the level that:

(1) Induces an appropriate production response;

(2) Induces a weakening of strict environmental regulations; or

(3) Reduces global consumption through an economic contraction.

One would clearly <u>not</u> advocate options (2) or (3); these options are solely presented to show what the levers are that would bring the oil market back into balance.

Given that none of these responses have yet occurred during a near quadrupling of oil prices (in dollars) over the past six years, it is an open question of how high oil prices (denominated in dollars) will go.

It may be difficult for institutional investors to access energy-futures relative-value strategies because of the lack of scalability in the back-end of both the oil curve and other energy derivatives curves. Perhaps these strategies will need to be limited to generating alpha on top of a structural exposure to commodities.

According to UBS (2008), the stock prices of the major US oil companies are still discounting only a $58-per-barrel price for oil. This discount could provide energy hedge funds, which specialize in petroleum-complex stocks, a structural edge.

Also, what might be some of the implications of a structurally higher-priced energy market? One consequence may be reduced reliance on trucking in the U.S. It is notable that Warren Buffet started investing in railroad companies last year.

The beneficiaries of the oil boom have obviously been the oil-exporting countries. What are they investing in? It is noteworthy that starting in the Spring of 2007, as the sovereign wealth funds became more formally organized in investing their dollar windfalls, that the turning points in oil prices have mirrored the fortunes of the Euro, which would be consistent with the well-telegraphed currency-diversification activities by oil-exporting countries.

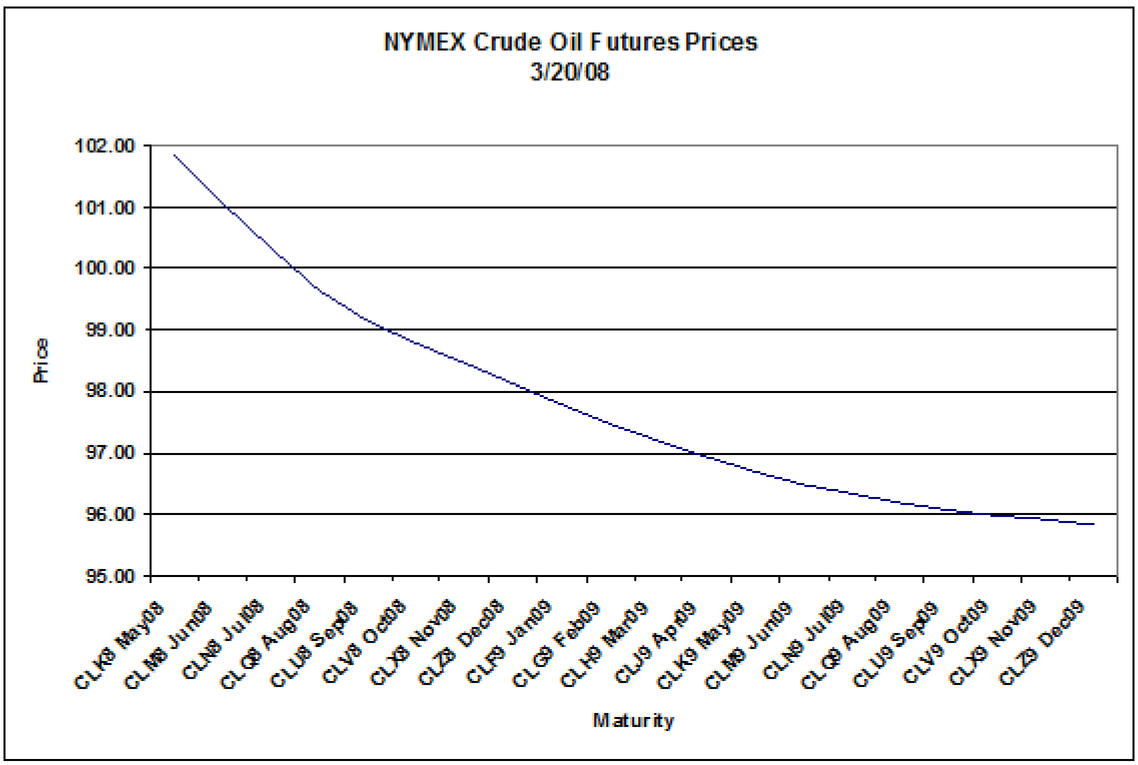
Now in examining the various choices of how to invest in the natural-resource markets, one's decision is made simplest when an investment has the following characteristics:

- positive carry;
- transparency; and
- scalability.

As of the writing of this paper, this is precisely the state of the front-month oil futures contract at this time, as illustrated in Figure 2.

Figure 2

NYMEX Crude Oil Futures Curve



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